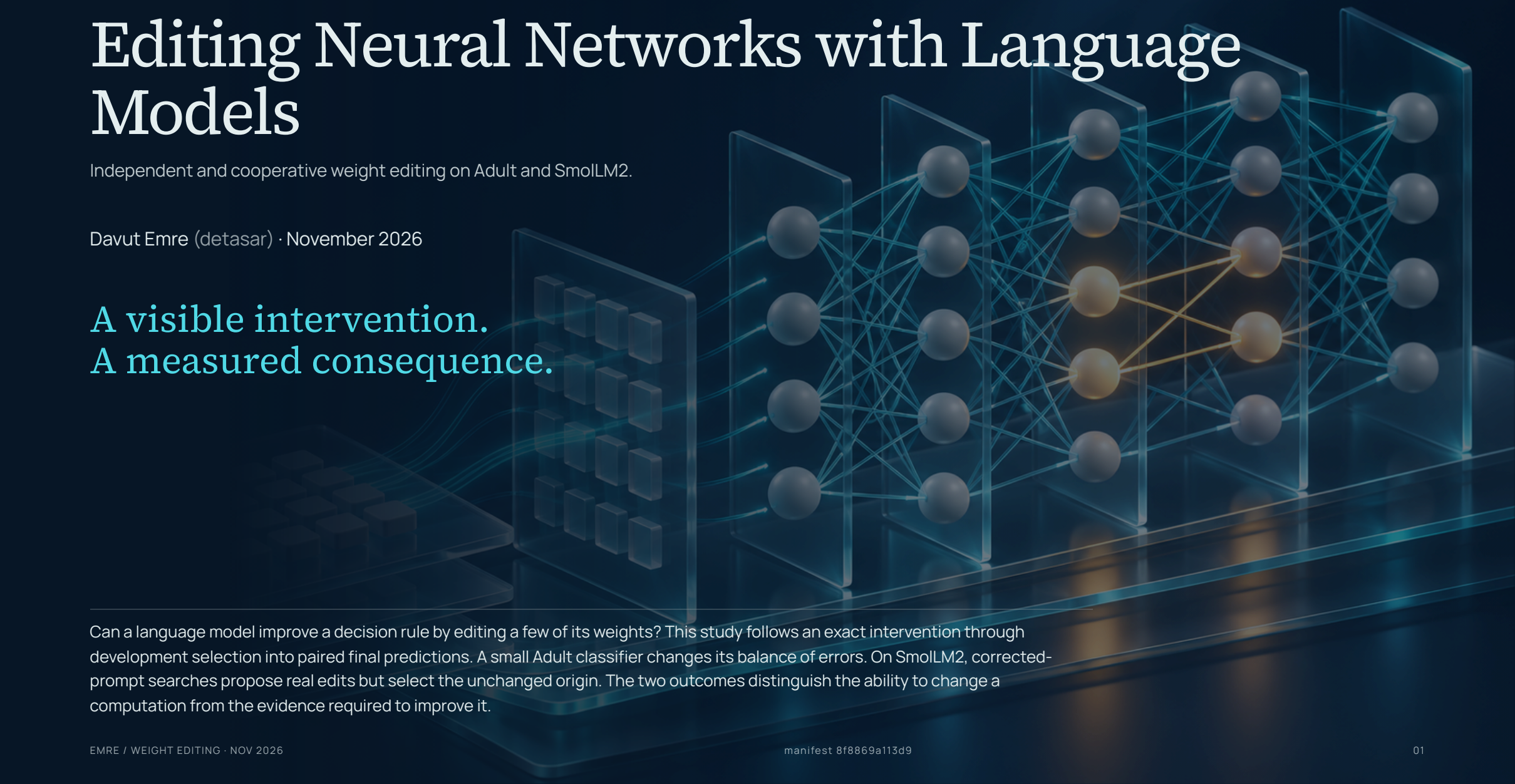
Editing Neural Networks with Language Models

Independent and cooperative weight editing on Adult and SmolLM2.

Davut Emre (detasar) · November 2026

A visible intervention.
A measured consequence.



Can a language model improve a decision rule by editing a few of its weights? This study follows an exact intervention through development selection into paired final predictions. A small Adult classifier changes its balance of errors. On SmolLM2, corrected-prompt searches propose real edits but select the unchanged origin. The two outcomes distinguish the ability to change a computation from the evidence required to improve it.

STUDY MAP

Two scales, one question

A small classifier makes the intervention legible. A language model tests selection at scale.

104

Adult

Encoded predictors
Binary income classification
Permanent lesion; surviving fc2 edits

×

24

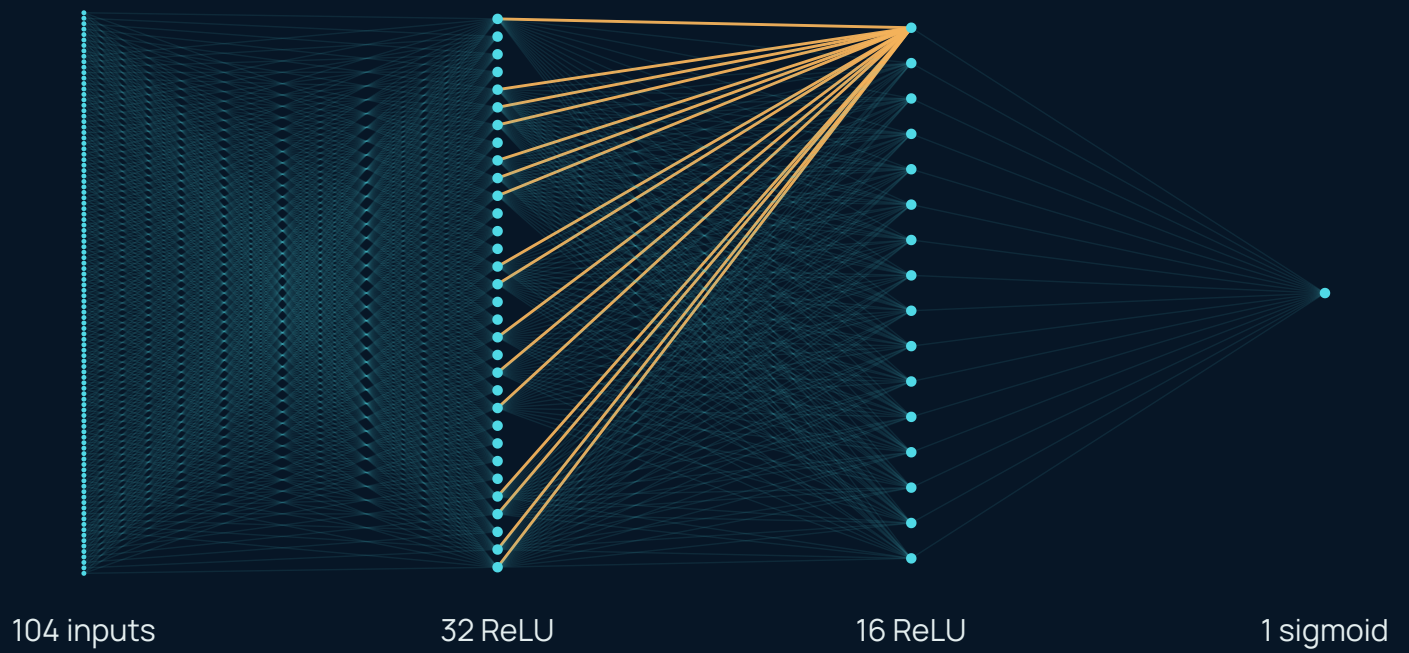
SmolLM2 blocks

Base language model
ARC-Easy + PIQA
Bounded down_proj edits

The study is inspired by Takehi, Teoh, Zhang and Khattab's Sparks of Weight-Space Programming (2 November 2026). Our independent test uses a damaged Adult classifier and an intact SmolLM2-1.7B base model, with different tasks and intervention domains from that article. Adult makes each changed scalar inspectable; SmolLM2 tests bounded editing at language-model scale. The endpoint comparison is exploratory and confined to one seed101 session. Read the reference article.

The whole Adult network

104 inputs → 32 ReLU → 16 ReLU → 1 sigmoid probability.



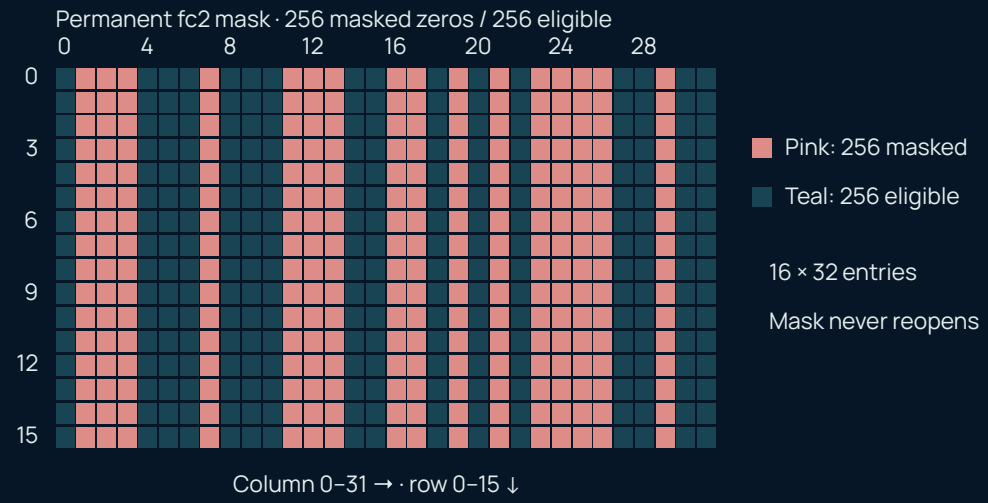
All 153 nodes · first-layer edges sampled · actual fc2 mask and support. Flow pulse is explanatory.

Adult maps 104 encoded predictors through hidden layers of 32 and 16 ReLU units to a sigmoid income probability. Sex and race are excluded from the encoded predictors. The figure includes all 153 nodes; only first-layer edges are sampled. The released graph supplies actual weights, the permanent mask and the executed selected patches. A changed edge denotes support of an intervention, rather than a measured activation or a causal attribution.

PERMANENT LESION

The mask stays closed

Removed connections remain zero; edits can only change surviving fc2 entries.



$$W_d = W_{clean} \odot (1 - M), \qquad \Delta \odot M = 0$$

M: 16×32 binary mask; 1 means permanently removed. ⊙: elementwise product.

Wd: damaged origin from Wclean. Δ⊙M = 0 keeps every masked scalar zero.

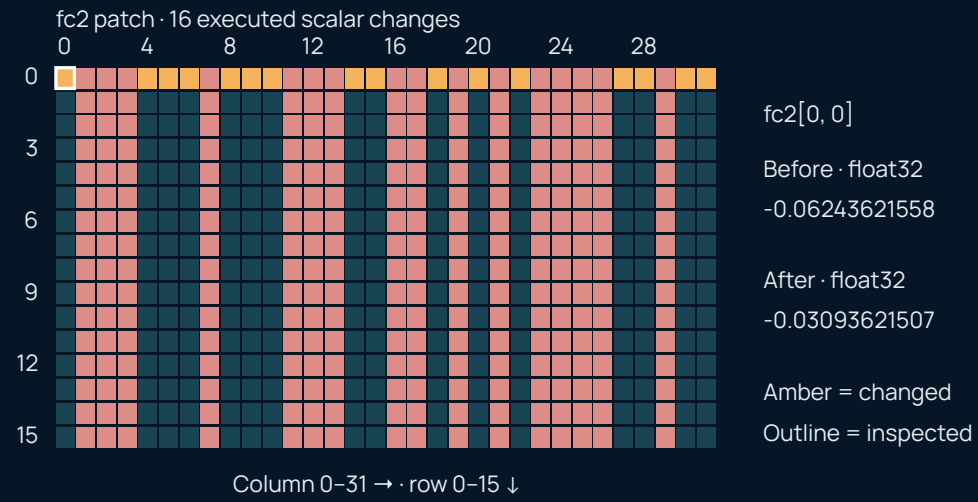
‖Δ‖F: square root of summed squared executed changes; relative norm divides by origin norm (limit .05).

The lesion zeros fixed fc2 columns. The 16×32 matrix displays all 256 masked zeros and 256 eligible entries; the mask excludes removed entries from every legal patch. Each candidate starts from the immutable damaged origin, so an editor cannot accumulate changes by submitting a sequence. The calibration damage criterion was insufficient. On the final panel the intact classifier has lower balanced accuracy but higher ordinary accuracy, ROC-AUC and probability quality: the relevant interpretation is an error-balance tradeoff.

A REAL SCALAR

Before. After. Exactly.

The visual support and the scalar inspector refer to the same executed patch.



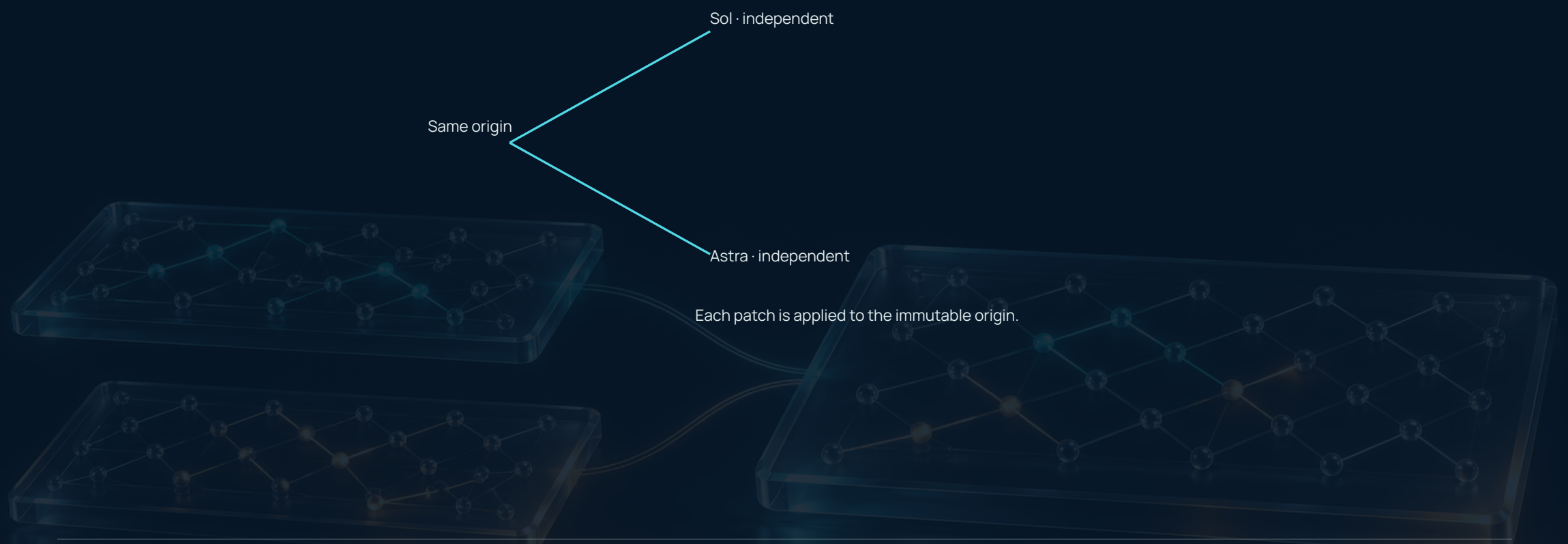
EXECUTED ADULT PATCH · SOL

Scalar fc2[0, 0] -0.06243621558 → -0.03093621507 16 changed · L2 0.143815856 · relative 0.048917676

The inspector reads the exported executed float32 values before and after the selected Adult patch. Sparse deltas affect surviving fc2 entries only. Changed scalar counts and L2 norms describe actual post-cast differences. Selecting a development candidate reconstructs its scalar assignment from the same stored origin; the frozen selected endpoint uses the directly exported values. This makes a visible edge traceable to a numerical operation.

Two branches from one origin

Sol and Astra never receive one another's source attempts.



Sol and Astra each begin an independent source search from the same origin. Neither sees the other's source attempts. The original Adult allowance is ten candidates per solo search; the joint search has five attempts and additional development history. These are comparisons of selected endpoints under that design, rather than a controlled estimate of equal-compute cooperative synergy.

THE ACTION LOOP

Propose → validate → measure

Development feedback guides edits. Final feedback never chooses the patch.

01 **Inspect** → 02 **Propose JSON** → 03 **Validate** → 04 **DEV feedback**

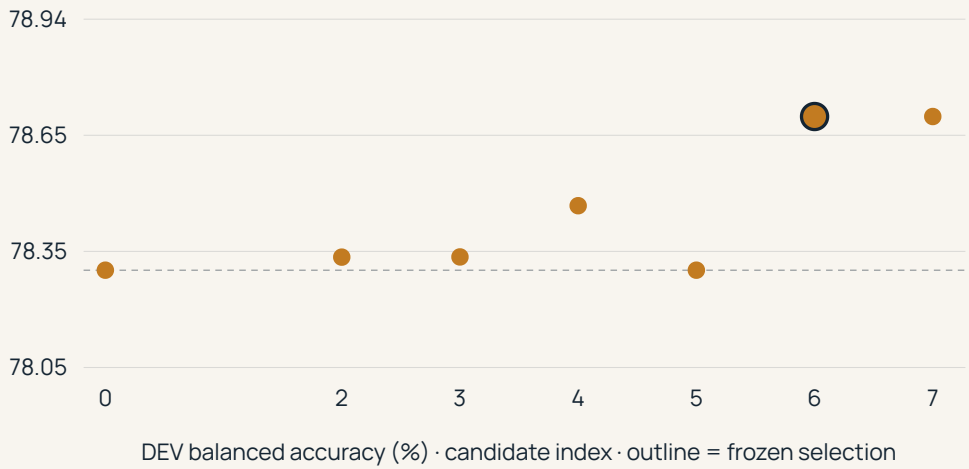
Freeze the chosen patch. Then measure FINAL.

Adult prompt · LM v2 prompt · Action schema

The controller separates inspection, structured JSON proposals, legal validation and development feedback. Adult changes only surviving fc2 weights; SmolLM2 changes down_proj tensors with a global changed-scalar fraction at most .001 and per-tensor relative Frobenius change at most .05. Invalid attempts receive no score. Prompts, schemas and runtime revisions are linked in the technical appendix. No gradients are used to produce these edits.

Selection is part of the result

A higher primary development score wins; equal scores prefer the smaller edit.



Sol · candidate 6 · 16 changed scalars · L2 0.14381586 · DEV 78.694638%

$$\arg\min_i(-d_i, \|\Delta_i\|_2, \#\Delta_i, i)$$

i: actual candidate index; di: measured DEV balanced accuracy (LM uses acc_norm).

Minimize the tuple from left to right: −di first means maximize the DEV score.

Ties prefer smaller L2 norm $\|\Delta i\|_2$, then fewer changed scalars #Δi, then earlier i. Candidate 0 is eligible.

Candidate zero is an eligible unchanged origin. Selection minimizes the tuple of negative development primary score, edit L2 norm, changed-scalar count and candidate index. Adult uses balanced accuracy; the language model uses character-normalized accuracy. Dots show measured candidate scores at actual indices, with no interpolated measurements. The outlined point is the frozen selection; final outcomes never participate in selection or stopping.

Share experience, keep the origin

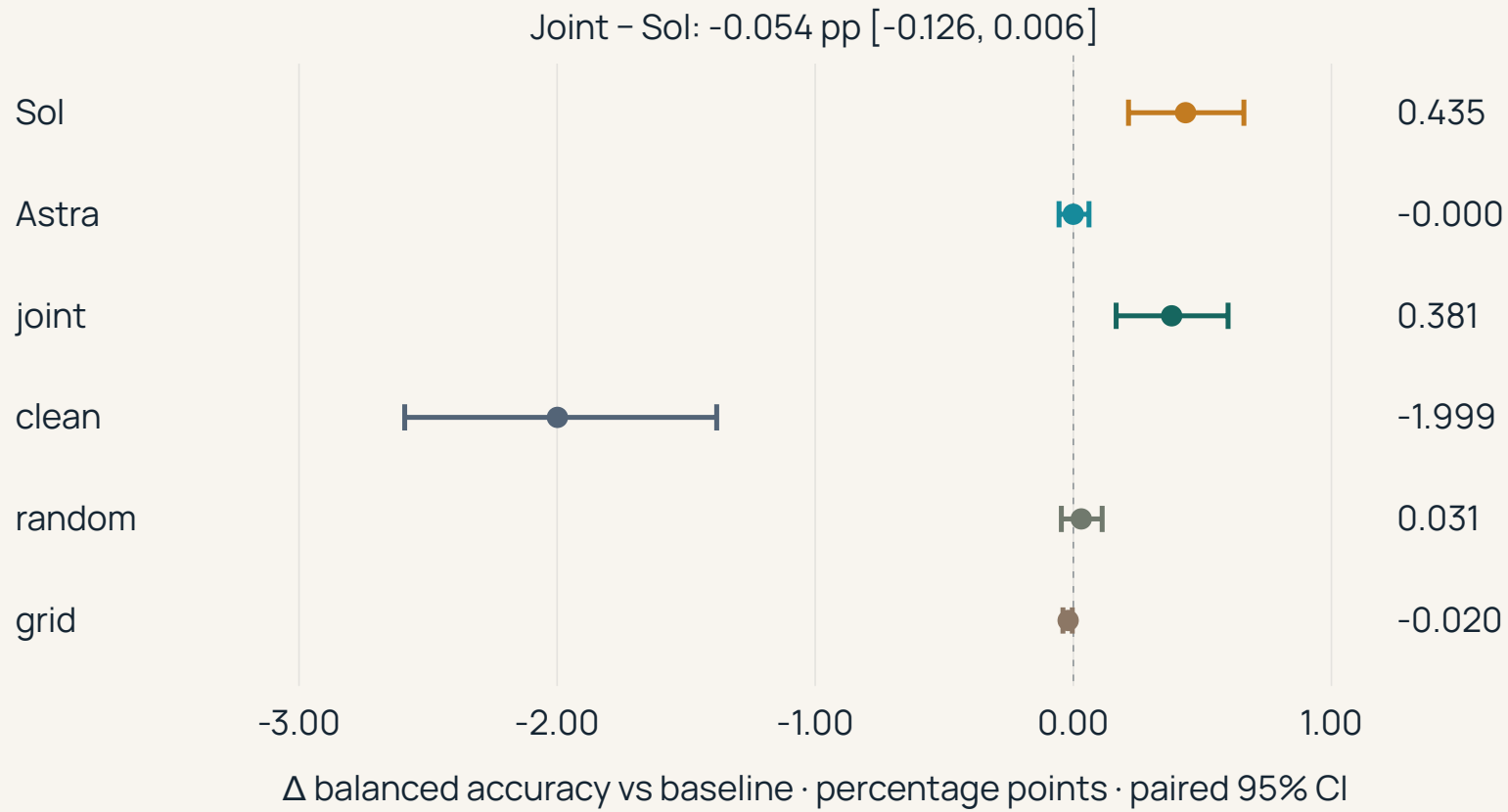
The joint search receives development history, not a serial sum of solo edits.



The joint session receives the independent source searches' development experience. It proposes a complete patch against the same origin; its result is never the serial sum of Sol and Astra. This separates information sharing from weight accumulation. The held-out paired comparison with Sol is shown with the final forest plot, allowing the cooperative endpoint to be assessed without assuming that cooperation improves it.

A small shift in balanced accuracy

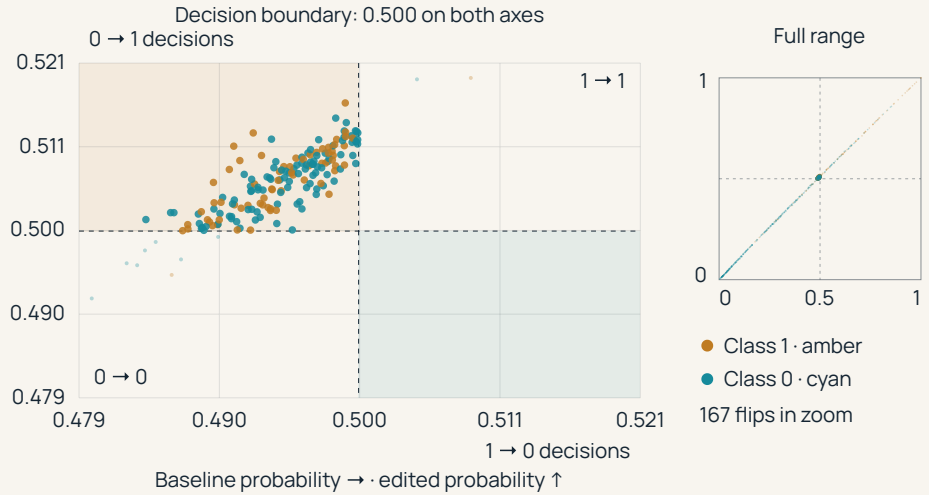
Sol and joint move the primary metric upward; Astra is effectively flat.



The common axis reports percentage-point differences in held-out balanced accuracy. Sol and joint exceed the damaged origin; Astra is effectively flat. Sol and joint also exceed the random and grid controls in the paired comparisons. Intervals use 2,000 class-stratified paired bootstrap draws with seed101. They quantify uncertainty across examples within this session, rather than variability between independent editor sessions.

Who changes sides?

Positive-class rescues come with negative-class regressions.



Class 1 · n=3,846

+65 rescues / **−0** regressions

Class 0 · n=12,435

+0 rescues / **−102** regressions

FINAL · Sol: 65 positive rescues / 0 regressions; 0 negative rescues / 102 regressions. BA gives each class half the weight; accuracy counts each row equally.

$$BA = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right)$$

TP / (TP+FN): positive recall; TN / (TN+FP): negative recall.

BA averages the two recalls equally, regardless of class size.

A rescue changes wrong → correct; a regression changes correct → wrong on the same saved row.

Balanced accuracy gives equal weight to the two label classes, whereas ordinary accuracy gives equal weight to examples. On this panel the negative class is larger. Positive-class rescues can therefore raise balanced accuracy while more negative-class regressions lower total correct decisions. The threshold zoom includes every decision flip and a deterministic sample of unchanged rows within its bounds. The inset retains the full probability range, using anonymous ordinals rather than original feature records.

Better balance is not recovery

Clean has lower BA, but better accuracy, ranking and probability quality.

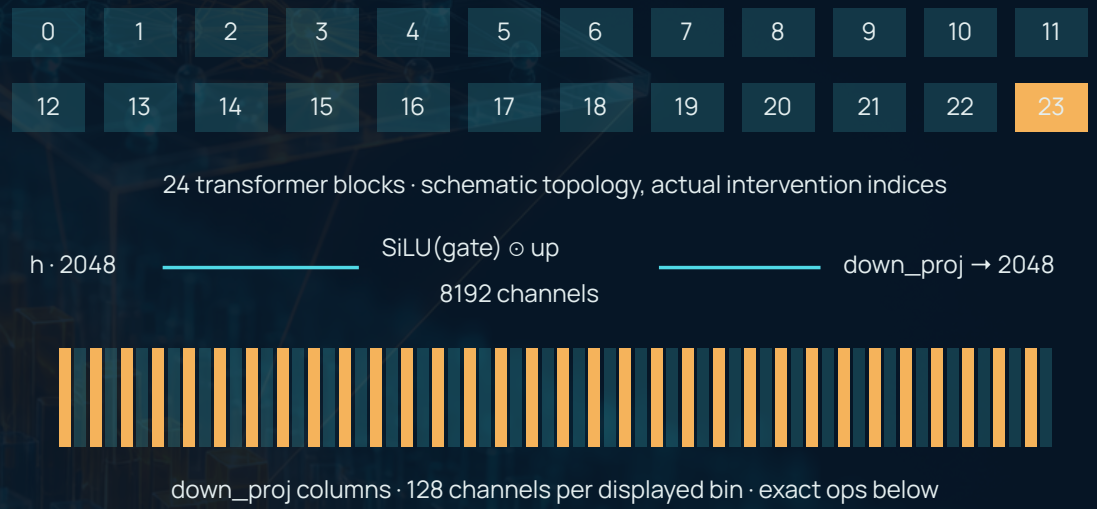
Adult final · 16,281 paired examples

State	BA ↑	Accuracy ↑	AUC ↑	Log-loss ↓	Brier ↓
baseline	78.586%	82.857%	0.889869	0.378285	0.120487
Sol	79.021%	82.630%	0.889981	0.382725	0.122237
Astra	78.586%	82.802%	0.889885	0.378581	0.120608
joint	78.967%	82.630%	0.889975	0.382383	0.122106
clean	76.587%	85.634%	0.907734	0.314733	0.100082
random	78.616%	82.808%	0.889981	0.37868	0.120655
grid	78.566%	82.827%	0.889862	0.376637	0.120089

The intact reference is strongest on accuracy, ranking and calibration while its balanced accuracy is below the damaged origin. Sol and joint increase the primary metric but lose ordinary accuracy and probability quality relative to damaged. Random and grid supplements provide non-editor comparisons under frozen states. Since clean minus damaged balanced accuracy is negative, a conventional balanced-accuracy recovery fraction is undefined for this session.

Edit a channel contribution

2048 → 8192 → 2048, with selected columns scaled or sparse scalars changed.



Sol · candidate 1 · 65,536 changed scalars · L2 14.45690302 · DEV 75.000000%

$$u = \mathrm{SiLU}(W_g h) \odot (W_u h), \qquad y = W_d u$$

h: 2048 residual inputs; Wg and Wu map to 8192 intermediate channels.

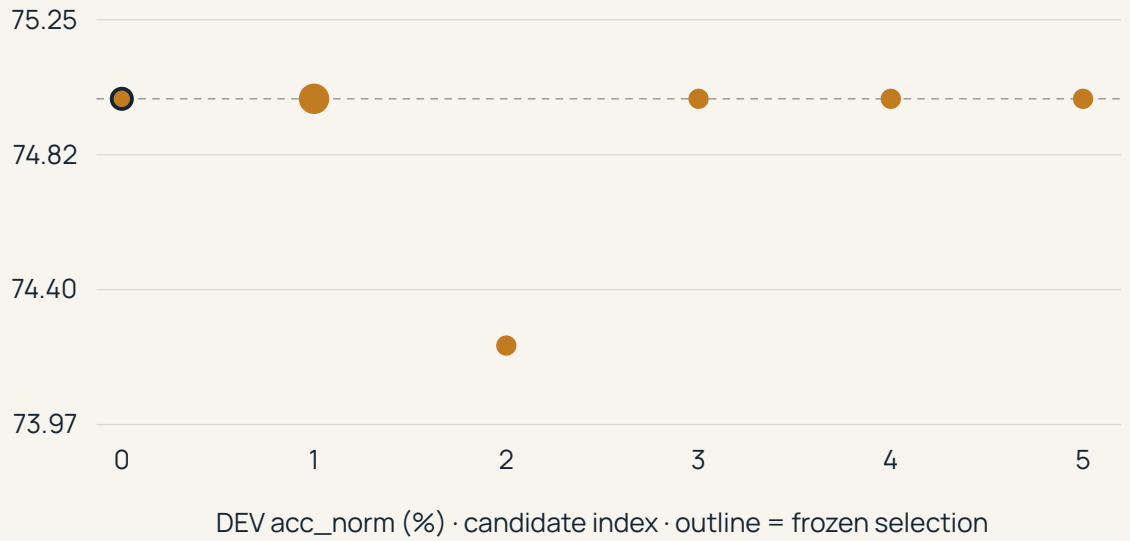
SiLU(z) = z σ(z); ⊙ multiplies gate and up branches component by component.

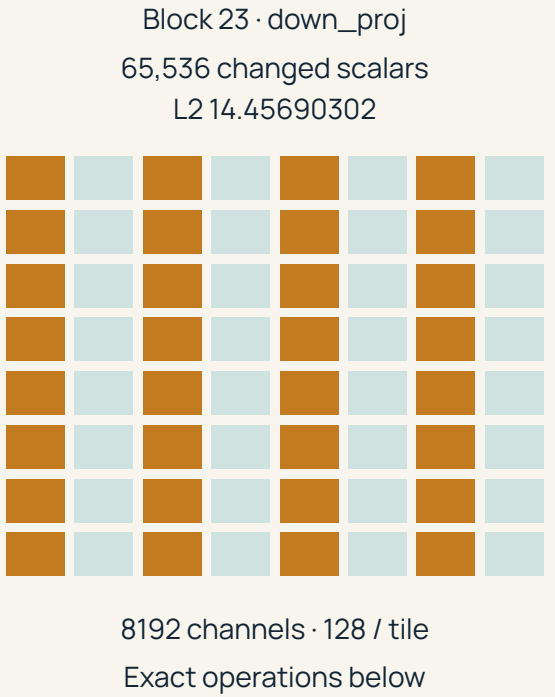
u: 8192 gated channels. Wd (down_proj): 2048×8192; y: 2048 outputs.

SmolLM2 has 24 transformer blocks. In each gated MLP, SiLU(gate) is multiplied elementwise by the up branch across 8,192 intermediate channels, and down_proj maps back to 2,048 dimensions. Scaling a down_proj column changes a channel contribution. The display aggregates 128 channels per bin, while tensor names, exact columns, factors and post-cast norms remain available in the operation inspector.

A real search can select no change

Nonempty candidates were measured. The unchanged origin still wins.





Sol · candidate 1 · 65,536 changed scalars · L2 14.45690302 · DEV 75.000000%

The corrected float32 v2 measurement is complete. Sol and Astra each have five nonempty source candidates. Joint has five attempts: attempt 1 is invalid and unscored; valid indices are 0, 2, 3, 4 and 5, including the unchanged origin and a final no-op. Every main condition selects candidate zero. The unchanged candidate's zero norm resolves ties in its favor, making selection of no intervention a measured outcome.

One state. Two benchmarks.

Four labels share one executed weight identity and one measured endpoint.

ARC-Easy

73.437500%

acc_norm · 376 / 512

Raw acc 77.343750%

396 / 512 correct

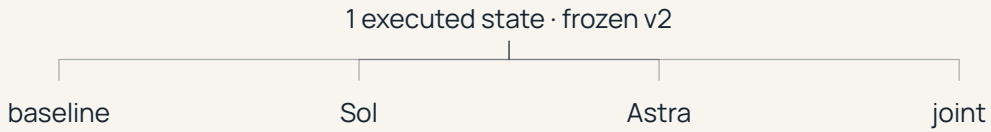
PIQA

78.515625%

acc_norm · 201 / 256

Raw acc 80.078125%

205 / 256 correct



$$s(c \mid q) = \frac{\sum_t \log p(c_t \mid q, c_{<t})}{\mathrm{len}_{chars}(c)}$$

q: question; c: choice text; ct: its continuation token at position t.

Sum log p(ct I q,c<t): continuation log likelihood; divide by character count, not token count.

Choose the first maximum normalized score; acc_norm is the fraction of correct choices. Raw accuracy omits the divisor.

These endpoints are the newly measured v2 results on ARC-Easy and PIQA. Normalized accuracy divides each choice continuation log likelihood by the character length of the choice text; raw accuracy uses the unnormalized likelihood. Baseline, Sol, Astra and joint share the exact executed weight identity, so their final predictions are reused as one state. Their paired differences, rescue/regression counts and confidence intervals are exactly zero.

Intervention, decision, selection

Adult changes decisions. LM selection rejects change. Both are results.

01 A real weight intervention can change a decision.

02 Adult gains balanced accuracy, with a tradeoff.

03 SmolLM2 v2 selects the unchanged origin.

The experiment connects three levels: a legal numerical intervention, a changed decision and a selection policy. Adult demonstrates a small balanced-accuracy improvement with a visible class tradeoff. SmolLM2 demonstrates that real proposed edits may fail to displace an eligible unchanged origin. Broader claims require independent sessions and controlled allowances. Saved numeric evidence, exact prompts and canonical scoring rules make both outcomes inspectable without a new model call.

Technical appendix

From tensor to prediction

Adult forward pass

$$h_1 = \mathrm{ReLU}(W_1 x + b_1),\ h_2 = \mathrm{ReLU}(W_2 h_1 + b_2),\ p = \sigma(W_3 h_2 + b_3)$$

W_1:32×104; W_2:16×32; W_3:1×16. Biases match output dimensions.

Legal changes

$$W' = W_d + \Delta, \quad \Delta \odot M = 0, \quad \frac{\|W' - W\|_F}{\|W\|_F} \le .05$$

M marks removed Adult entries. LM per-tensor relative Frobenius ≤ .05, global changed fraction ≤ .001; no overlapping operations.

LM application

$$W'_{:,j} = W_{:,j} + \mathrm{cast}_{dtype}(W_{:,j}(s-1))$$

s is the requested scale; sparse changes add cast(delta) to a scalar. Counts and norms use actual post-cast changes.

Choice selection

$$\hat{c}_{norm} = \arg\max_c \frac{\sum_t \log p(c_t \mid q, c_{<t})}{\mathrm{len}_{chars}(c)}$$

Raw acc omits the character divisor. Native first-argmax tie handling applies.

Metrics & paired uncertainty

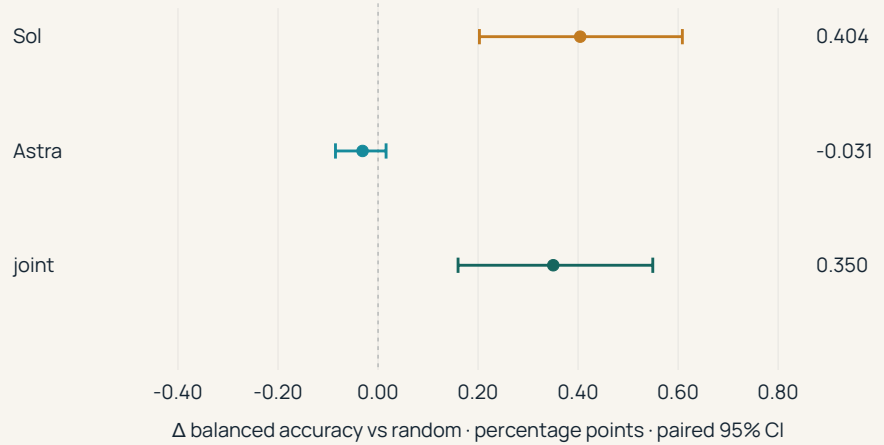
$$BA = \frac{1}{2}(\mathrm{TPR} + \mathrm{TNR}), \quad Acc = \frac{TP + TN}{N}$$

$$LL = -\frac{1}{N}\sum_i [y_i \log p_i + (1 - y_i)\log(1 - p_i)], \quad Brier = \frac{1}{N}\sum_i (p_i - y_i)^2$$

$$\delta_b = BA(y[I_b], p_A[I_b]) - BA(y[I_b], p_B[I_b])$$

The same row indices I_β are sampled in both conditions, with replacement separately within each Adult label class. Seed 101; 2000 draws; percentile 2.5/97.5%. Positive = income >50K (3,846); negative = ≤50K (12,435). ROC-AUC describes ranking; lower log-loss and Brier are better probability quality.

A rescue changes wrong to correct; a regression changes correct to wrong. Intervals concern paired examples in one session, not editor-session variation. The insufficient-damage flag is true; no BA recovery denominator is reported.



All final metrics

Adult final · 16,281 paired examples

State	BA ↑	Accuracy ↑	AUC ↑	Log-loss ↓	Brier ↓
baseline	78.586%	82.857%	0.889869	0.378285	0.120487
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Raw acc 80.078125%

205 / 256 correct



All four LM labels share these v2 values and one weight identity. Paired acc/acc_norm differences vs baseline are zero, with 95% intervals [0,0] and zero rescues/regressions.

Methods, history & sources

One exploratory seed101. Adult solos start independently, ten allowed candidates each; joint has five attempts and sees only development history. All patches start at the same origin. Final feedback never selects or stops the search. Clean/random/grid supplements and LM v2 measurements are complete.

v1 LM prompt said float16 while execution used float32. v2 corrects that wording and uses fresh source sessions, but the analyst had seen v1 final results and the panels are reused. This is not an independent replication. Both historical v1 and main v2 selected the unchanged origin; scores remain separately versioned.

MPS float32 train8 was an engineering sensitivity check. No float32 train64 gate was run; this does not establish full-benchmark numerical equivalence. Exporting the snapshot made no model calls; original v2 measurement did complete.

Prior work: Rikiya Takehi, Jayden Teoh, Alex Zhang and Omar Khattab, Sparks of Weight-Space Programming, 2 November 2026. This independent study uses different targets and intervention domains.

Full methods · Versions and revisions · Reproduction · Source provenance

Exact prompts: v1 LM · v2 LM · Adult · Joint. Exact scorer and task continuation conventions: scorer reference.

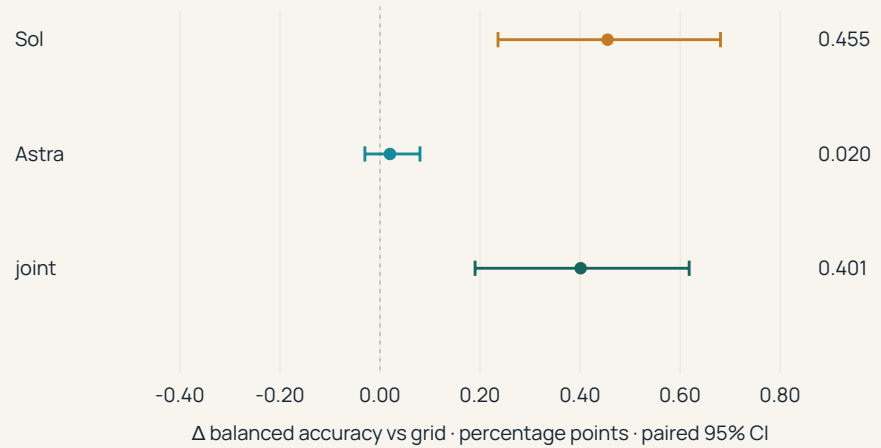
Manifest identity: 8f8869a113d93000bb8fffd547258b36ab955a6bf7a7f8ddbe589fbdbd231d38

Work & complete comparisons

Recorded target work · native units kept separate

Phase	Examples	Forwards	Seconds	Editor calls
LM v2 DEV Sol	672	2,690	260.046	7
LM v2 DEV Astra	672	2,690	262.152	7
LM v2 DEV joint	416	1,666	161.753	14
LM v2 FINAL · unique state	768	2,558	248	0
Adult controls · CPU batches	48,843	3	0.005182	0

LM forwards are per-choice requests; Adult forwards are batched calls. One LM final state feeds four method labels; it is one observation, not four replications.



Why balance and accuracy diverge

Let R_k and G_k count rescues and regressions in class k, and n_k be its fixed size. Recall changes by $(R_k - G_k)/n_k$. Substituting this into balanced accuracy gives:

$$\Delta BA = \frac{1}{2}\left(\frac{R_1 - G_1}{n_1} + \frac{R_0 - G_0}{n_0}\right), \quad \Delta Acc = \frac{R_1 + R_0 - G_1 - G_0}{n_1 + n_0}$$

For Sol, positive net correct = 65 / 3,846, negative net correct = -102 / 12,435. Thus ΔBA = 0.434901 percentage points while Δaccuracy = -0.227259 points. A negative-class regression costs less BA per example because that class is larger.

From a scalar to its local contribution

$$z_2' - z_2 = \Delta h_1, \qquad y' - y = (W'_{down} - W_{down})u$$

Adult h_1 is unchanged because fc1 is fixed. The preactivation difference enters ReLU, then the final linear layer and sigmoid. For the LM expression, u is the same local MLP input: the equation explains the edited projection, not all downstream logits. Exact post-cast differences, rather than requested decimals, define Δ. The intervention norm is the square root of the sum of squared executed scalar differences; the relative norm divides it by the origin tensor norm.